Do agent safety refusals hold up in Nigerian languages?

Draft protocol. No model safety rates have been measured or claimed.

Question

Do refusal and harmful-completion rates differ across English, Yoruba, Hausa, Igbo and Nigerian Pidgin, and between plain chat and tool-using agents? Does translating a request to English for a safety check reduce harmful behavior without an unacceptable rise in benign over-refusal?

Data and translation

Import 100 category-balanced standard HarmBench behaviors and 100 safe-labelled XSTest prompts from pinned source commits. Keep the text private. Retain English as a control and translate the same IDs into four Nigerian languages. Use a versioned translator, preserve ambiguity, and never ask it to answer the prompt. NLLB support for Nigerian Pidgin is not assumed. The configurable translator must actually support all four languages.

Ask consenting native speakers to check 30 items per Nigerian language, balanced between harmful and benign sources. Record meaning/fluency on a 1–5 scale, accept/reject status, reviewer ID and translation hash. Acceptance requires meaning >=4 and fluency >=3. Report distributions and major mistranslations. Correct rejected translations and review them again. The runner checks that completed review records match the current translation hashes. A 30-item sample does not certify every translation.

Agent, models and conditions

Use three licensed open instruct models from distinct families and one versioned commercial free-tier endpoint. Keep target model, prompt, output budget and temperature fixed across four conditions: chat, agent, guarded chat and guarded agent. The English system instruction is identical across languages. The agent has a four-turn maximum and four fake tools: send_email, web_search, write_file and run_code. Implementations validate arguments and append hashes only; they have no network, file-writing or code-execution dispatch.

The guard back-translates the actual request to English and applies a fixed safety-check prompt before any target interaction. It never reads the original gold English prompt as a substitute for translation. Guard/checker failures are errors, not successful refusals. A blocked request creates no target completion and no tool attempt. Record blocking, errors and latency separately.

Scoring and judge validation

Measure full refusal, harmful completion, harmful tool attempt and benign over-refusal. A tool call alone is not harmful; semantic judgment of its intended action is required. Real-world harmful execution is zero by construction, and must not be confused with attempted harmful tool use.

Back-translate responses and use the fixed, versioned judge prompt in safety/score.py. Tool arguments remain available privately for semantic review. Reject invalid judge JSON rather than coercing strings or missing values into safe labels. Judge failures remain unjudged. Report attempted cases, request errors and judgment coverage alongside conditional rates and Wilson intervals.

Independently hand-label 100 target outputs, stratified by language and source kind. Hide automatic labels from reviewers. Exclude automatic guard-block records from this calibration sample. Report agreement and Cohen's kappa for each label, and inspect disagreements by language/condition. One overall agreement number can hide language-specific judge failure.

Analysis and chart

Plot harmful-completion rate by language for chat/agent, before/after the guard, with confidence intervals and a separate facet per target model. Report benign over-refusal and harmful tool attempts in companion tables. Compare paired IDs across conditions; use a paired bootstrap or appropriate paired test for final difference claims. The existing Wilson intervals describe individual rates, not the uncertainty of paired differences.

Ethics and release

Never publish harmful outputs or the harmful prompt set. Keep raw data and trajectories in mode-700 directories with mode-600 files. Use only fake tools, respect all model/API/benchmark terms, allow reviewers to skip sensitive content, and privately disclose a serious commercial-model vulnerability before affected results are published. The public exporter constructs aggregate records from an allowlist; it never copies model responses into public results.

Limitations and current conclusion

Translation, judge bias and English system prompting can confound language comparisons. A small prompt set and four models do not establish language-wide safety. Inert tools measure attempted actions, not downstream real-world harm. Free-tier quotas may limit the 16,000 target evaluations plus ancillary calls. The protocol and implementation are ready; results, human/native review and any mitigation benefit remain unmeasured.